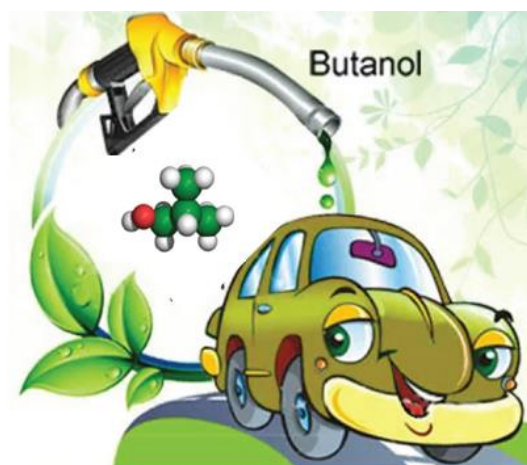


LECTURE-19 (BIOBUTANOL)

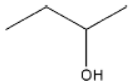


➤ INTRODUCTION

- Biobutanol is a 4-carbon alcohol (butyl alcohol, butanol) produced from the same feedstock source as bioethanol which include corn grain and other biomass.
- When we talk of the term 'biobutanol', means it has been produced by microbial source using any substrate or from biomass feedstocks.
- Butanol is colorless and flammable.
- It can be mixed with ethanol, ether and other organic solvent.
- It can be used as a solvent, in cosmetics, hydraulic fluids, detergent formulations, drugs, antibiotics, hormones and vitamins, as a chemical intermediate in the production of butyl acrylate and methacrylate, and additionally as an extract agent in the manufacture of pharmaceuticals.
- It's molecular formula is C_4H_9OH and the carbon atoms can form either a straight-chain or a branched structure, resulting in isomers with distinct properties.
- The isomers are based on the location of the – OH group and carbon chain structure.
- The different structures, properties and main applications have been shown in **Table-1**.
- Although the properties of butanol isomers are somewhat different in octane number, boiling point, viscosity, etc., the main applications are similar in some aspects, such as being used as solvents, industrial cleaners, or gasoline additives.
- All these butanol isomers can be produced from fossil fuels by different methods, only *n*-butanol, a straight-chain molecule structure can be produced from biomass.
- In the present lecture, biobutanol produced from biomass feedstock is the topic of discussion.
- It is known that bioethanol as one of the biofuel has been applied in automobiles with gasoline in different blending proportions.
- Biobutanol is one of the new types of biofuel.
- Due to its superior properties over other biofuel, it has continuously attracted the attention of researchers and industrialists throughout the world.

- Now biobutanol is seen as a potential option for fuel additive or a direct replacement for gasoline or diesel.

Table 1. Structures, Properties and Main Applications of N-Butanol, 2-Butanol, Iso-Butanol and Tert-Butanol

	n-Butanol	2-Butanol	iso-Butanol	tert-Butanol
Molecular structure				
Density (g/cm ³)	0.81	0.806	0.802	0.789
Boiling point(°C)	118	99.5	108	82.4
Melting point(°C)	-90	-115	-108	25-26
Refractive index(n _D 20)	1.399	1.3978	1.3959	1.3878
Flash point(°C)	35	22-27	28	11
Motor octane number	78	32	94	89
Main applications	Solvents-for paints, resins, dyes, etc. Plasticizers- improve a plastic material processes Chemical intermediate -for butyl esters or butyl ethers, etc. Cosmetics- including eye makeup, lipsticks, etc. Gasoline additive	Solvent Chemical intermediate- for butanone, etc. Industrial cleaners -paint removers Perfumes or in artificial flavors	Solvent and additive for paint Gasoline additive Industrial cleaners -paint removers Ink ingredient	Solvent Denaturant for ethanol Industrial cleaners- paint removers Gasoline additive for octane booster and oxygenate Intermediate for MTBE, ETBE, TBHP, etc.

- Fuel properties of conventional gasoline and biofuels used in the transportation sector are compared in Table-2.

Table-2: Properties of Gasoline and Biofuels used in Transportation.

Fuel property	Gasoline	(Bio)Butanol	(Bio)Ethanol	Biodiesel	Reference
Chemical structure	C ₄ to C ₁₂	CH ₃ CH ₂ CH ₂ CH ₂ OH	CH ₃ CH ₂ OH	Methyl esters of C ₁₂ to C ₂₂ and fatty acids	(USDOE 2013)
Carbon ratio (%)	86.3	64.9	52.2	77.6	(GREET 2012)
Density (g L ⁻¹)	744.7	809.7	789.3	887.9	(GREET 2012)
Feedstock	Crude oil	Sugars from e.g. corn, grains or agricultural waste, (cellulose)	Sugars from e.g. corn, grains or agricultural waste, (cellulose)	Fats and oils from e.g. animal fats, soy beans, cooking oil or rapeseed	(USDOE 2013)
Energy content (lower heating value, MJ kg ⁻¹)	32.4	27.8	21.3	33.3	(GREET 2012)
RON ¹	91–99	96	129	-	(Lee <i>et al.</i> 2008a)
MON ²	81–89	78	102	-	(Lee <i>et al.</i> 2008a)

¹Research octane number, ²Motor octane number.

➤ ADVANTAGES OF BUTANOL AS FUEL

- Except the use of solvent, chemical intermediate and extract agent, butanol also can be used as fuel, which attracted people's attention in recent years.
- Because of the good properties of high heating value, high viscosity, low volatility, high hydrophobicity, less corrosive, butanol has the potential of a good fuel.
- The properties of butanol and other fuels are compared in **Table-3**.

Table-3: Properties of Butanol and other Fuels

Fuel	Octane number	Cetane number	Evaporation heat (MJ/kg)	Combustion energy(MJ/dm ³)(%vol)	Flammability limits	Saturation pressure (kPa) at 38°C
Gasoline	80-99	0-10	0.36	32	0.6-0.8	31.01
Methanol	111	3	1.2	16	6-36.5	31.69
Ethanol	108	8	0.92	19.6	4.3-19	13.8
Butanol	96	25	0.43	29.2	1.4-11.2	2.27

- When ethanol is mixed with gasoline (less than 10%), there exists some disadvantages.
- Firstly, the heating value of ethanol is much less than gasoline.
- The fuel consumption increases if the engine is not retrofitted.
- Secondly, acetic acid is produced during the burning process of ethanol, which is corrosive to the engine.
- The preservative must be added when the ethanol proportion upper than 15%.
- Thirdly, ethanol is hydroscopic and the liquid phase separation may be occurring with high water proportion.
- Furthermore, ethanol as fuel cannot be preserved easily and it is more difficult in the process of allocation, storage, transition than that of gasoline.

- Compared with ethanol, butanol overcomes above disadvantages and it shows potential advantages.
- Butanol has higher energy content and higher burning efficiency, which can be used for longer distance.
- The air to fuel ratio and the energy content of butanol are closer to gasoline.
- Therefore, it can be easily mixed with gasoline in any proportion.
- It is less volatile and explosive, has higher flash point, and lower vapor pressure, which makes it safer to handle and can be shipped through existing fuel pipelines.
- In addition, butanol can be used directly or blended with gasoline or diesel without any vehicle retrofit.
- Actually, the first-time synthesis of biobutanol at laboratory level was reported by Pasteur in 1861 and the industrial synthesis of biobutanol was started during 1912–1914 by fermentation.
- However, before 2005, butanol was mainly used as solvent and precursor of other chemicals due to the product inhibition and low butanol productivity.
- To bring awareness to butanol's potential as a renewable fuel, David Ramey drove his family car from Ohio to California on 100% butanol.
- And then, two giant companies DuPont and BP declared to finance development of a modernize production plant supported by research and development.
- The economy of biobutanol production also was revaluated.
- The research of a continuous fermentation pilot plant operating in Austria in the 1990s introduced new technologies and proved economic feasibility with agricultural waste potatoes.

➤ PRODUCTION METHODS OF BUTANOL

Butanol can be obtained using chemical technologies, such as Oxo-synthesis and aldol condensation. It is also possible to produce butanol in the process of fermentation by bacteria and butanol as one of the products called biobutanol. The most popular bacteria species used for fermentation is *Clostridium acetobutylicum*. Because the main products of this process containing acetone, butanol and ethanol, the fermentation is called ABE fermentation.

• Chemical Process

- ✓ Butanol can be produced by chemical synthesis.
- ✓ One process is Oxo-synthesis, which involves the reaction of propylene with carbon monoxide and hydrogen in the presence of cobalt or rhodium as the catalyst.
- ✓ The mixture of n-butyraldehyde and isobutyraldehyde are obtained and then the mixture can be hydrogenated to the corresponding n-butanol and isobutyl alcohols .
- ✓ The reactions are as following:

$$\text{CH}_3\text{CH}=\text{CH}_2 + \text{CO} + \text{H}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CHO} + (\text{CH}_3)_2\text{CHCHO} \quad (1)$$

$$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO} + \text{H}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \quad (2)$$

$$(\text{CH}_3)_2\text{CHCHO} + \text{H}_2 \rightarrow (\text{CH}_3)_2\text{CHCH}_2\text{OH} \quad (3)$$
- ✓ When using cobalt as the catalyst, the reaction proceeds at 10~20MPa and 130~160 °C, the products ratio of n-butyraldehyde and isobutyraldehyde is 3.

- ✓ Rhodium as the catalyst used in industry since 1976 and the reaction processes at 0.7-3MPa and 80-120 °C.
- ✓ The products ratio of *n*-butyraldehyde and isobutyraldehyde can reach 8-16.
- ✓ Another route is aldol condensation, which involves the reaction of condensation and dehydration from two molecules of acetic aldehyde.
- ✓ And then, the product crotonaldehyde is transformed into *n*-butanol by hydrogenation at 180 °C and 0.2 MPa.
- ✓ The reaction is as following:

$$\text{CH}_3\text{CH}=\text{CHCHO} + 2\text{H}_2 \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} \quad (4)$$
- ✓ Comparing the two processes, Oxo-synthesis route has the advantages of materials easily obtained, comparable moderate reaction conditions, enhanced ratio of *n*-butanol to isobutyl alcohol.
- ✓ So, Oxo-synthesis process is the main industrial route for *n*-butanol production.
- ✓ **Note:** There are also some other fossil oil derived raw materials such as ethylene, propylene and tri-ethyl-aluminium or carbon monoxide and hydrogen which are used in butanol production.

• Biological Process

- ✓ Butanol can also be obtained from biological ways with the renewable resources by the microorganism through fermentation.
- ✓ The *Clostridia* genus is very common for butanol synthesis under anaerobic conditions, and the fermentation products are often the mixture of butanol, acetone and ethanol.
- ✓ The production of butanol by fermentation using lignocellulosic feedstocks is also known as ABE fermentation, i.e. acetone-butanol-ethanol fermentation.
- ✓ Compared with the chemical method, biological route has the distinct advantages.
- ✓ For example, it can utilize the renewable resources such as wheat straw, corn core, switch grass, etc. as feedstocks.
- ✓ Biological process has high product selectivity, high security, less by-products.
- ✓ Furthermore, the fermentation condition of butanol production is milder than that of chemical process and the products are easier to separate.
- ✓ **History of Biological Process for Manufacturing of Biobutanol**
 - Biological production of butanol under anaerobic conditions is typically referred to in literature as a part of 'ABE fermentation', since acetone, butanol and ethanol are usually produced simultaneously in this process.
 - Louis Pasteur was the first to report about microbial butanol production in 1862.
 - However, the first production utilizing the Weizmann process began only in 1913, aiming to produce acetone for rubber synthesis.
 - Later in 1916, the first industrial-scale ABE fermentation began operation due to a high demand for acetone during World War I, and after the armistice in November 1918, most of the plants were shut down.

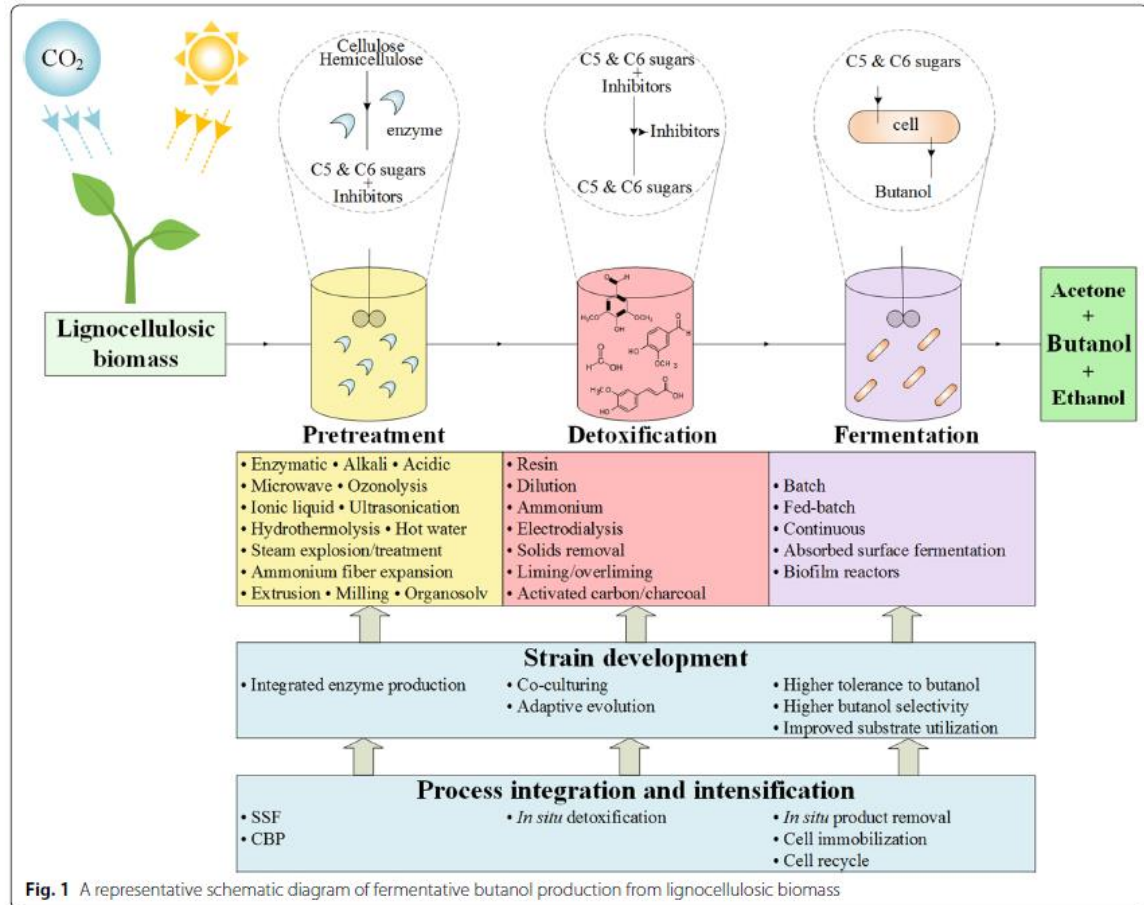
- Industrial ABE fermentation, however, kept expanding worldwide, facilitated by the usability of butanol as solvent.
- In 1945, two-thirds of the butanol and one tenth of the acetone in the U.S. were produced by ABE fermentation processes.
- However, their share in the total output declined rapidly during the 1950s mainly because of the acute competition with the expanding petrochemical industry and decreasing feedstock availability.
- ABE fermentation became popular again in the 1970s after the oil crisis, and it has since been gaining increasing interest owing to the advancements in Metabolic Flux Analysis, Metabolic Engineering, Gene KO Homologous Recombination, and Complete Genome Sequencing, holding promise of improved production yields and productivities for more economic microbial production processes.
- There are several excellent reviews covering the historical development of ABE fermentation in detail.
- The historical details of ABE fermentation has also been briefed in **Table-4**, given on the next page.

Table-4: History and Development of ABE Fermentation.

Years	Scientists	Contributions and Developments
1861	Louis pasteur	First reported the butanol production from microbial source and termed as ' <i>Vibron Butyrique</i> '.
1905	Schardinger	Production of acetone by fermentation
1910	Perkins and Weizmann	Reported the best metabolic pathway for isoprene or butadiene production from isoamyl alcohol or butanol
1911	Fernbach	Isolated microbial population that could ferment potatoes to butanol, but not maize starch.
1912	Weizmann	Butanol or isoamyl alcohol formation through fermentation found to be important for the success of synthetic rubber process.
1912-1914	Weizmann	During this, multiple numbers of cultures reported, one from these BY. Later this was named as <i>Clostridium acetobutylicum</i> . This strain possessed ability to convert various starchy biomass into acetone and butanol.
1913	Strange and Graham	Strange and Graham Ltd. Applied for a patent of a process using Fernbach's bacillus strain. By this year they started production of acetone and butanol at Rainham, using potatoes as substrate.
1916	Weizmann	British Admiralty took over the Strange and Graham Ltd. plant at King's Lynn under "The Defence of the Realm Act". They made changes to Weizmann process and started utilizing maize as basic material for the process.
1918	Weizmann	During this time acetone production for cordite and airplane dope started and in Canada butanol manufacturing and storage in vats also started. Conversion of butanol to methyl ethyl ketone begins in small quantity.
1919-1920	Weizmann and EloiRicard	In 1920 U.S. Weizmann patent allotted them a license to start butanol manufacturing at the Terre Haute plant. They started operation according to U.S license and later got worldwide process rights of this patent.
1923	Terre Haute plant, with Weizmann	This was the time when bacteriophage contamination affected the yield badly. To rectify this problem a research team developed at the Terre Haute plant under the supervision of Weizmann.
1935	J. Hastings	During this time the Commercial Solvents Corp and the Distiller's Co. in a joint venture project developed plant that could produce industrial level of alcohol, acetone and buanol with distillation units. The plant was completely operated by trained British personnel under guidance of J.Hastings. The phase immunized <i>C.saccharoacetobutylicum</i> strain used for the process.
1936-1941		New Acetone Butanol fermentation plants constructed in Philadelphia, Pa., Baltimore, Md., and Puerto Rico and became functional with the expire of Weizmann patent in 1936. After this year many new isolated strains reported with better fermentation capability from molasses. During this period about 18 patents issued covering process using different strains, adapted by big production houses like Publiker Industrial Inc., Commercial Solvents Corp., Western Condensing Co., and U.S. Industrial Chemicals Co.
1936-1960		Plants construction for the solvent production also started in different part of world such as South Africa, Australia, Japan, and India. In between 1950 and 1960 various researches about AB fermentation process published and implemented for acetone and butanol production by continuous process. With these modification operation started in Dokshukino in the USSR in 1960.
1950-1960		After the last phase of second world war acetone-butanol fermentation rapidly retarded in different part of the world, may be due to rise in substrate cost and increased competition to the industry for cheap raw substrate material.
1970-1982		By 1970 Organic chemical Factories of Egyptian Sugar and Distillation Company was partially functional. South African plant, the only plant working until 1982.
1980-1990		During this period intensive research focused in a direction of process improvement, process optimization, strain improvement, improvement of culture cultivation, using various alternative fermentation substrates and improvement of product removal technologies.
1990 onwards		With the development of recombinant DNA technology, genetic manipulation and strain improvement of solvent-producing clpstridia successfully completed in laboratories. These approaches are proper documented in various research articles, patents and reviews. These are well implemented at industrial scale production.

➤ FERMENTATIVE BUTANOL PRODUCTION FROM LIGNOCELLULOSIC BIOMASS

A typical conversion process from lignocellulosic biomass to butanol involves three major steps: pretreatment, detoxification and fermentation. A representative schematic diagram of the process is shown in **Figure-1**.

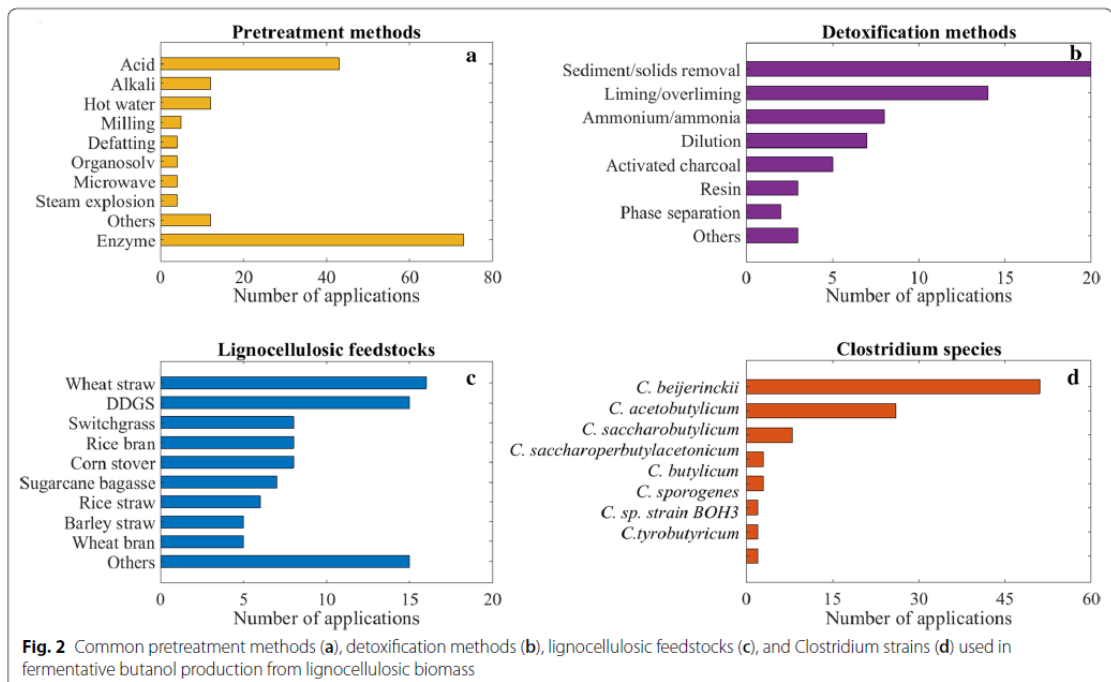


• Pretreatment

- ✓ Lignocellulosic biomass is a favorable feedstock since it is the most abundant renewable biomass resource on the planet, and, compared to sugars from e.g. sugar cane or maize, it avoids direct fuel-versus-food competition.
- ✓ Its main constituents are cellulose, hemicellulose and lignin.
- ✓ The opening of the lignocellulosic biomass structure and the release of sugar content from hemicellulose and cellulose with other cross-linked units and the residual non-hydrolyzed raw feedstock is called pretreatment.
- ✓ Conversion of biomass into its main constituents is referred to in literature as fractionation, which is sometimes used interchangeably with pretreatment, i.e. pretreatment is mentioned as a way of achieving biomass fractionation, or the term fractionation is used as (part of) a pretreatment method.
- ✓ In the present study, for simplicity reasons we name all steps involved in the conversion of the feedstock to sugars as pretreatment though enzymatic hydrolysis of the polysaccharide fractions is often referred to as a step that is distinct from other pretreatment measures.

- ✓ Predominance of enzymatic hydrolysis in the pre-treatment methods in **Figure-2 (a)** shows its widespread application to produce fermentable sugars from lignocellulosic biomass.
- ✓ Milling/grinding, extrusion, microwave and ultrasonication are common physical pretreatment methods that open up the physical structure of lignocellulosic biomass.
- ✓ Physico-chemical methods such as steam explosion, steam treatment, hydrothermolysis, ammonium fiber expansion, and hot water treatment cause both the structure to unravel and a release of sugar monomers and dimers.
- ✓ Major chemical pretreatment methods are alkali, acidic, ozonolysis, ionic liquid and organosolv treatments.
- ✓ Enzymatic hydrolysis using suitable enzyme mixtures degrades polysaccharides such as cellulose and xylan to fermentable C6 and C5 sugar monomers, respectively.
- ✓ Typically, combinations of several of the above-mentioned pretreatment methods are employed depending on the feedstock.
- ✓ Operating conditions of pretreatment are crucial since a small change in the operating parameters can cause great differences in reduced sugar composition and concentration as well as inhibitory compounds, consequently negatively affecting enzymatic hydrolysis, fermentability and the cost of substrate.
- ✓ Therefore, it is crucial to examine the feasibility of any pretreatment method with respect to the generation of inhibitors, energy consumption, operating cost, and sugar yield.

Figure-2: Common Pretreatment Methods (a), Detoxification Methods (b), Lignocellulosic Feedstocks (c), and *Clostridium* Strains (d) Used in Fermentative Butanol Production From Lignocellulosic Biomass



- **Detoxification**

- ✓ Compounds that are inhibitory to microorganisms and enzymes are often generated during pretreatment.
- ✓ Cellulose and hemicellulose should ideally only yield sugar monomers such as glucose, xylose, and mannose.
- ✓ However, severity of some pretreatment conditions converts those sugars into furfural, 5-hydroxymethyl furfural (HMF), formic acid, acetic acid, levulinic acid and salts, which can be inhibitory.
- ✓ Partial decomposition of lignin generates inhibitory (poly)phenolic aromatic compounds such as *p*-coumaric acid, ferulic acid, syringe aldehyde, vanillic acid and vanillin.
- ✓ Contrary to ethanol-producing microorganisms like the yeast *Saccharomyces cerevisiae*, furfural, HMF or acetic acid are not inhibitory to clostridial butanol producers at relatively low concentrations, rather they are reported to be stimulatory.
- ✓ Another common compound generated during pretreatment of lignocellulosic biomass is formic acid.
- ✓ It is found to be inhibitory to *C. acetobutylicum* at 0.5 g/l and 0.074 g/l (1 mM) inside the cell wall due to acid crash.
- ✓ Therefore, if larger amounts of inhibitors are present after pretreatment, it is a necessity to remove these for a successful fermentation.
- ✓ For this purpose, several detoxification methods such as electrodialysis, liming/overliming, activated carbon/charcoal, dilution, and resin treatments as depicted in **Figure-2 (b)** are applied.
- ✓ Even though it is not specifically mentioned as a detoxification method, solid/sediment removal by filtration or centrifugation is also commonly applied to alleviate the inhibitory effects of the solids and undissolved lignin in the lignocellulosic hydrolysates.
- ✓ It is important to note that the enzymes used in the hydrolysis step can be inhibited by the compounds mentioned above as well as their sugar yields, which can impose a limit to high substrate concentration.
- ✓ Alternative lines of research currently target new pretreatment methods that are less prone to inhibitor formation (like organosolv or other low-temperature methods) and thus ideally do not require detoxification prior to fermentation, as well as increasing the inhibitor tolerance of fermentation strains e.g. by means of adaptive evolution.

- **Fermentation**

- ✓ ABE fermentation is biphasic; first, acetic acid and butyric acid are produced in the acidogenesis phase, then the acids are re-assimilated to yield the solvents acetone, butanol and ethanol.
- ✓ Batch fermentation is the most widely used method due to simple operation and low risk of contamination.
- ✓ Low cell density can result in low productivity, and absorbed substrate fermentation and biofilm reactors have been applied to overcome this problem in batch processes.
- ✓ Fed-batch mode is beneficial to tackle substrate inhibition by gradually adding the substrate, thus keeping the substrate concentration below toxic levels.

- ✓ However, fed-batch fermentation should still be accompanied by in situ product removal to alleviate product inhibition.
- ✓ Continuous fermentation (chemostat) has advantages over batch and fed-batch modes such as improved productivity.
- ✓ Multi-stage, immobilized cell, cell recycling and bleeding techniques have been applied to improve chemostat performance.
- **Strain Development**
 - ✓ Strain development refers to any modifications in the butanol production strain done by random mutagenesis and selection, like in adaptive laboratory evolution, or directed, rational and/or systems biology guided genetic modification employing metabolic engineering and synthetic biology to improve fermentation performance nby means of increased tolerance to toxic components, butanol selectivity and productivity, and improved substrate utilization and range.
 - ✓ In general, detoxification methods shown in **Figure-2 (b)** are used for removal of inhibitors present in the substrate and/or feedstock as described in “**Detoxification**” section.
 - ✓ Co-culturing with other species to eliminate toxic components such as oxygen in case of anaerobic fermentation is an alternative method.
 - ✓ Random mutagenesis and selection, and metabolic engineering have been applied for the same purpose.
 - ✓ Inhibition due to butanol accumulation is one of the greatest challenges.
 - ✓ Therefore, metabolic engineering and mutagenesis have been targeting this specific problem as well by developing strains with greater resistance to butanol toxicity.
 - ✓ A typical fermentative butanol production yields acetone and ethanol as well, which decreases the selectivity of the product of interest.
 - ✓ Metabolic engineering for disruption of acetone producing pathways, homo-butanol fermentation via chemical mutagenesis and metabolic engineering and conversion of acetone into isopropanol are among the strategies developed to address this issue.
 - ✓ Efficient utilization of the substrate is crucial to achieve a high butanol yield, thus improving fermentation performance.
 - ✓ Disrupting the genes responsible for *Carbon Catabolite Repression* and overexpression of genes responsible for xylose transport and catalytic enzymes (d-xylose isomerase, xylulokinase, and enzymes of PPP) are commonly followed approaches.
- **Process integration and intensification**
 - ✓ Process integration and intensification techniques are applied to obtain cost-effective fermentation processes.
 - ✓ Important process intensification approaches include (a) simultaneous saccharification and (co-) fermentation (SSF or SSCF) in which hydrolysis of polysaccharides present in (pre-treated) biomass is performed by externally produced and added hydrolytic enzyme mixes in situ with the simultaneous fermentation of the liberated sugars by a strain (or in the case of SSCF several strains with complementary sugar substrate spectrum) producing the product of choice, e.g. butanol, and (b) consolidated bioprocessing (CBP) in which the

saccharolytic enzymes are produced within the sugar fermenting culture e.g. by the target product producing strain itself or in co-culture with a partner strain specialized in enzyme production and secretion.

- ✓ Gas stripping, pervaporation, adsorption, liquid–liquid extraction, pertraction (membrane extraction), reverse osmosis and membrane distillation are in situ product removal methods used to alleviate inhibitory effects of butanol.
 - ✓ Fermentation with integrated gas stripping has widely been studied, mostly in fed-batch mode, which showed improved butanol productivity.
 - ✓ Cell immobilization and cell recycle are mostly integrated to fermenters operated in continuous mode to improve butanol productivity by preventing the loss of cell mass with the bleeding stream out from the fermenter.
 - ✓ Process integration and intensification measures therefore play crucial roles in optimizing butanol fermentation processes for improved performance and economic competitiveness.
- **Another Process Flow Diagram which can be alternatively used is given below:**

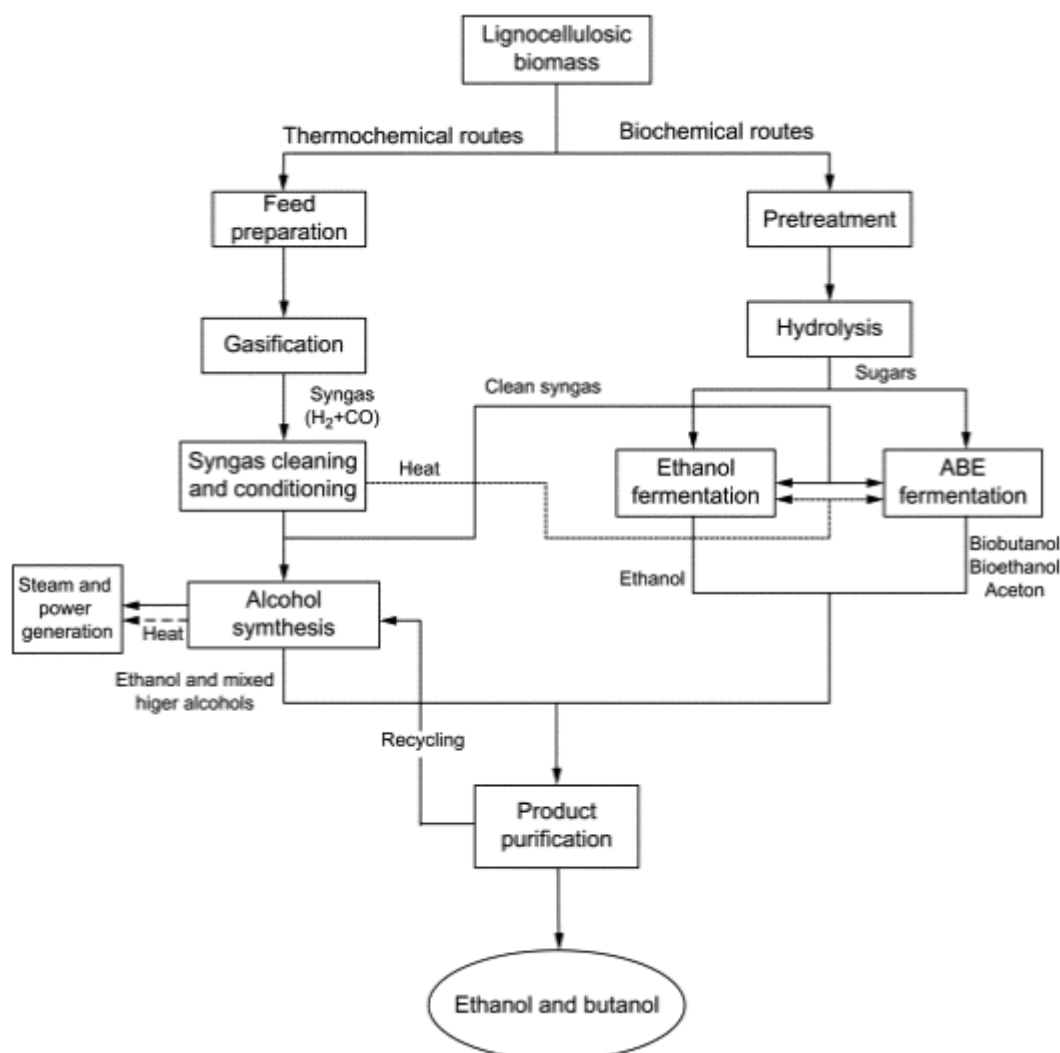


Figure-3: Manufacture of ethanol and butanol from linocellulosic feedstocks.

➤ CHALLENGES AND SOLUTION

- There are several challenges such as high substrate cost, solvent toxicity, low cell density and by-product formation that need to be addressed for sustainable and economical fermentative butanol production.
- These issues cause low butanol yield, titer, productivity and selectivity.
- Great efforts have been made to find cheap/free feedstock and cost efficient processing methods to overcome the high substrate cost problem, and several review papers address this issue in detail.
- Low solvent tolerance limits the butanol titer to maximum 2% dependent of the strain used, causing high downstream processing cost; therefore some reviews collected and discuss information on this specific challenge.
- Efficient separation of butanol from the fermentation mixture is another important topic with several reviews discussing particularly downstream processes for ABE fermentation.
- Strain improvement by metabolic engineering has an important role in optimizing butanol production.
- Main issues and possible solutions are summarized in **Table-5**.

Table-5: Challenges and Solution for Butanol Production by Fermentation

Challenges	Solutions
Increased operating cost due to high raw material cost	Switching to cheaper and sustainable raw material such as lignocellulosic materials
High cost of recovery due to low yield, limited sugar loading	Development of improved microorganisms with improved yield and/or butanol selectivity; Development of novel reactive separation processes
Capital and operating cost increases due to low volumetric productivity	Development of continuous fermentation processes for reduction in reaction time
Product recovery with conventional distillation is expensive and energy intensive	Development of low cost energy efficient methods for solvent purification and recovery.
High water usage is not sustainable and caused higher cost of effluent treatment.	Recycling the process water back through the fermentation.
Formation of undesirable or low value by-products	Engineering of respective knock-out or deletion strains
Culture degeneration	Avoid the excessive acidification of the culture.
Phage contamination	Good culture practices, proper sterilization, use of phage resistant strains

➤ GLOBAL SCENARIO

• **Global Bio-Butanol Market - Growth, Trends, and Forecast (2019 - 2024)**

- ✓ **Market Overview:** The market for bio-butanol is expected to register a compound annual growth rate (CAGR) of 8.36%, during the forecast period of 2019-2024. Major factors driving the market studied are the reduction of carbon emission to a great extent and gaining prominence as a building block for chemical manufacturing.

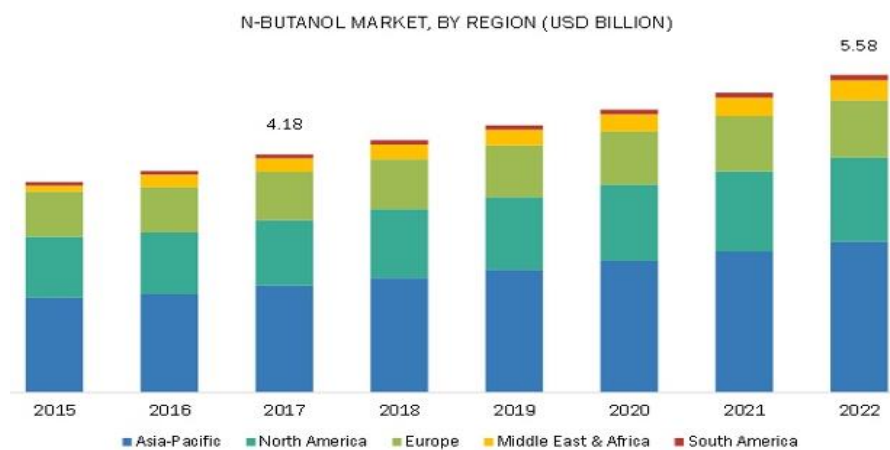
Bio-butanol is generating interest as a potential green alternative to petroleum fuels. It is preferred as a superior automobile fuel, as compared to bioethanol, owing to its high energy content. Furthermore, it is less corrosive in nature, and can be blended with any concentration with gasoline (petrol). Moreover, bio-butanol has low vapor pressure, thus lower volatility and evaporative emissions. Bio-butanol

also reduces greenhouse gas emissions, as it exhibits fewer emissions compared to other transport fuels. The United States and the European countries have started with fleet-testing of bio-butanol, in order to further promote the usage of bio-butanol. Growing number of cars on road and increasing sales of vehicles are expected to drive the market for motor fuel, which, in turn, may boost the market studied through the coming years.

The number of cars on the road is expected to rise by up to 2 billion, by 2050. According to OICA (Organisation Internationale des Constructeurs d'Automobiles, International Organization of Motor Vehicle Manufacturers) the total sales of all vehicles, including passenger and commercial, reached 96,804,390 million units, in 2017, from 93,905,634 million units, in 2016. This increase is leading to more usage of motor fuels in cars, which, in turn, may drive the market for bio-butanol as an alternative to transport fuel.

Asia-Pacific Region to Dominate the Market: In the Asia-Pacific region, China accounts for the largest market share in bio-butanol consumption. The country is the largest producer of paints and coatings in the Asia-Pacific region, with an estimated coating production of more than 15 million metric ton, which is expected to increase significantly in the near future. According to the China Adhesives and Tape Industry Association, the Chinese adhesive market grew by 7.5%, in terms of volume, and was valued at 7.887 million metric ton in 2017. The market grew further in 2018, and is expected to witness a positive trend through the forecast period. The increasing coatings, adhesive, and resin production is primarily supported by the growing construction, and infrastructural and automotive industries. China is promoting and undergoing urbanization, with a 60% target rate by 2020. The increased living spaces required in urban areas, resulting from urbanization, and the desire of middle-class urban residents to improve their living conditions are expected to have a profound effect on the housing market. Additionally, various paint manufacturers are expanding their product line, owing to the increasing demand for coatings. Major adhesive manufacturers are expanding their product line in India. The aforementioned factors are expected to drive the market growth, over the forecast period.

Competitive Landscape: The bio-butanol market is in the nascent stage and is apparently a consolidated one. Currently, Cathay Industrial Bio produces corn-based *n*-butanol for chemical applications at its biorefinery in Jilin Province, China. The company claimed to be the world's largest bio-butanol producer, based on active production capacity, in 2017. The potential entrants to the market include Microvi Biotechnologies, Cobalt Technologies, and ButaNext.



Source: Investor Presentation, Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis